SODIUM BICARBONATE

Hypertonic — extravasation

TCA overdose antidote

CHEMICAL NAME $NaHCO_3$ — 8.4% = 1 mmol/mL Na^+ and HCO_3^- (hypertonic, ~2,000 mOsm/L); 1.26% isotonic; oral tablets 500–600 mg

CLASS Systemic alkalinising agent (electrolyte)

TRADE NAMES Sodium bicarbonate 8.4% / 4.2% / 1.26% · Sodibic · Baking soda (oral)

THERAPEUTIC USES / INDICATIONS

- Severe metabolic acidosis (pH <7.1) in selected cases, especially with AKI.
- Tricyclic antidepressant overdose (QRS >100 ms, arrhythmias).
- Urinary alkalinisation (salicylate poisoning, rhabdomyolysis, methotrexate).
- Hyperkalaemia with acidosis; chronic metabolic acidosis in CKD (oral); prolonged cardiac arrest (not routine).

ADMINISTRATION

- **IV**: 8.4% via central or large vein (hypertonic, vesicant); 1.26% via peripheral line.
- Never mix with calcium (precipitate) or catecholamines (inactivated); flush line before/after.
- **PO** tablets for CKD acidosis.

DOSAGES

- **TCA toxicity:** 50–100 mL of 8.4% (1–2 mmol/kg) IV bolus, repeat to QRS <100 ms / pH 7.45–7.55.
- **Metabolic acidosis:** guided by base deficit: mmol = base deficit × weight (kg) × 0.2 (UK label); give half, then reassess with blood gases.
- **Urinary alkalinisation:** 1.26% 1 L over 2–3 h (target urine pH 7.5–8.5).
- **CKD acidosis (PO):** dose calculated individually from acid–base status (renal team).

PHARMACOKINETICS

- Dissociates to Na^+ and HCO_3^-; $HCO_3^- + H^+ \rightarrow CO_2 + H_2O$ (CO_2 exhaled — needs adequate ventilation).
- Excess bicarbonate excreted by kidneys.

PHARMACODYNAMICS

- Raises **plasma bicarbonate and pH**; alkalinises urine; sodium load and alkalaemia narrow QRS in TCA toxicity (Na^+ channel effect).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain.
- Belching, bloating (oral).

ADVERSE EFFECTS (SERIOUS)

- Metabolic alkalosis, hypokalaemia, hypocalcaemia (↓ ionised Ca^{2+} → tetany).
- Hypernatraemia, fluid overload.
- Paradoxical intracellular/CSF acidosis (CO_2 generation) in poor ventilation.
- Extravasation necrosis.

CONTRAINDICATIONS & CAUTIONS

- Metabolic or respiratory alkalosis; hypocalcaemia (relative).
- Hypoventilation (unless ventilated); hypernatraemia.
- Excessive chloride loss (vomiting).

KEY INTERACTIONS

- **Calcium salts:** precipitation.
- **Adrenaline, noradrenaline, dopamine:** inactivated — separate lines.
- **Lithium, salicylates:** ↑ excretion; **amphetamines, quinidine:** ↓ excretion.

NURSING MANAGEMENT

- Monitor ABG (pH, HCO_3^-, $PaCO_2$), K^+, Na^+, ionised calcium.
- ECG monitoring in TCA overdose (QRS).
- Check IV site; use central line for 8.4%.
- Monitor urine pH during alkalinisation.

PATIENT EDUCATION

- Report muscle twitching, cramps or swelling.
- Oral tablets: take as prescribed; may cause bloating.

KEY POINTS

- Alkalinising agent; 8.4% = 1 mmol/mL.
- Hypokalaemia, hypocalcaemia, alkalosis.
- TCA overdose: 1–2 mmol/kg to narrow QRS.
- Never with calcium or catecholamines.

SOURCES CHECKED **[968]** Sodium Bicarbonate 8.4% w/v solution for infusion (UK SmPC) **[448]** UKKA hyperkalaemia guideline (2023) *Full references p. 432–457*